DIGITAL MINE ON OPENGIZ

Ademix

3D open-pit operations twin

Live OpenEgiz twins connect mine events to dispatcher decisions.

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The load-haul-crush decision

A dispatcher must keep ore moving when equipment, road speed or crusher capacity changes.

01 **Observe**

Returned twin state shows trucks, queues, road condition, production and constraints.

02 **Decide**

The Decision workspace compares fixed and Smart Dispatch using matched model evidence.

03 **Act**

The operator restores the road or crusher, tows a failed truck, or parks surplus fleet.

FICTIONAL LAYOUT | SOURCE-CALIBRATED | SYNTHETIC TELEMETRY

OpenEgiz is the source of displayed state

The interface never reads the simulator directly. Every operational value returns through Eclipse Ditto.

Python simulation
Deterministic events

—

MQTT
Ditto Protocol

—

Eclipse Ditto
11 digital twins

—

Ademix API
Authenticated reads

—

3D command center
Returned state only

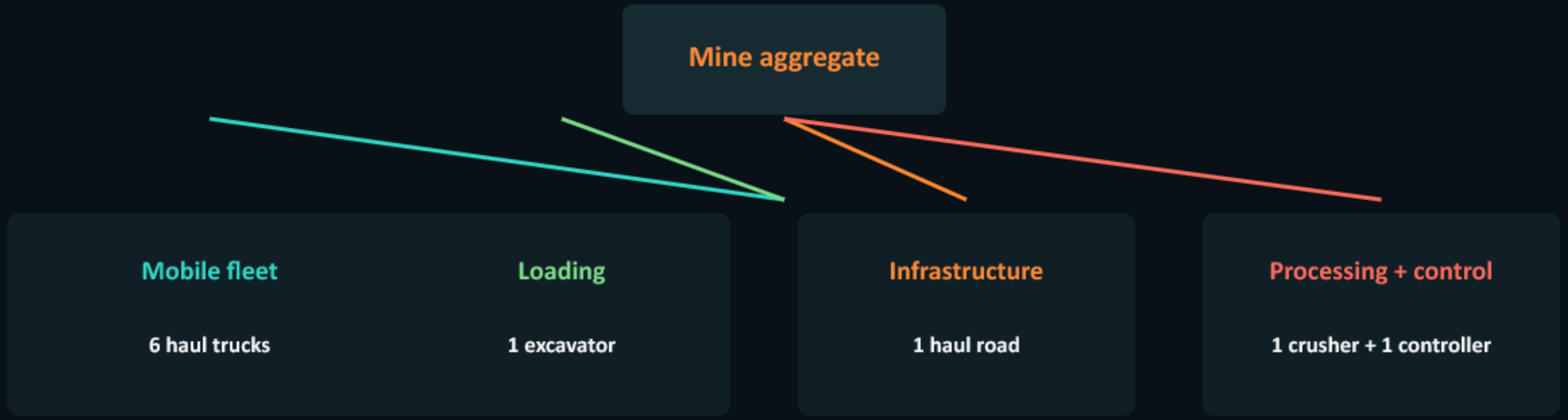
11
Ditto twins

143
required fields and units

20x
authoritative simulation clock

Eleven twins describe one production circuit

Thirty-eight explicit relationships make hierarchy, material flow, movement and control inspectable.



Relationships carry model meaning. Support buildings and two waste trucks remain visual context.

The mine view starts with the operating picture



Opening priorities

Production

Observed output and processed tonnes

Constraint

Current bottleneck and scenario

Fleet

Cycling and ready reserve trucks

Queue

One operator-relevant wait measure

Technical layers stay closed until requested.

Material balance and model assumptions

77 t
truck payload

770 t/h
modeled baseline

3.2 km
one-way route assumption

0 t
accepted mass-balance error

Material accounting

initial ROM + extracted ore

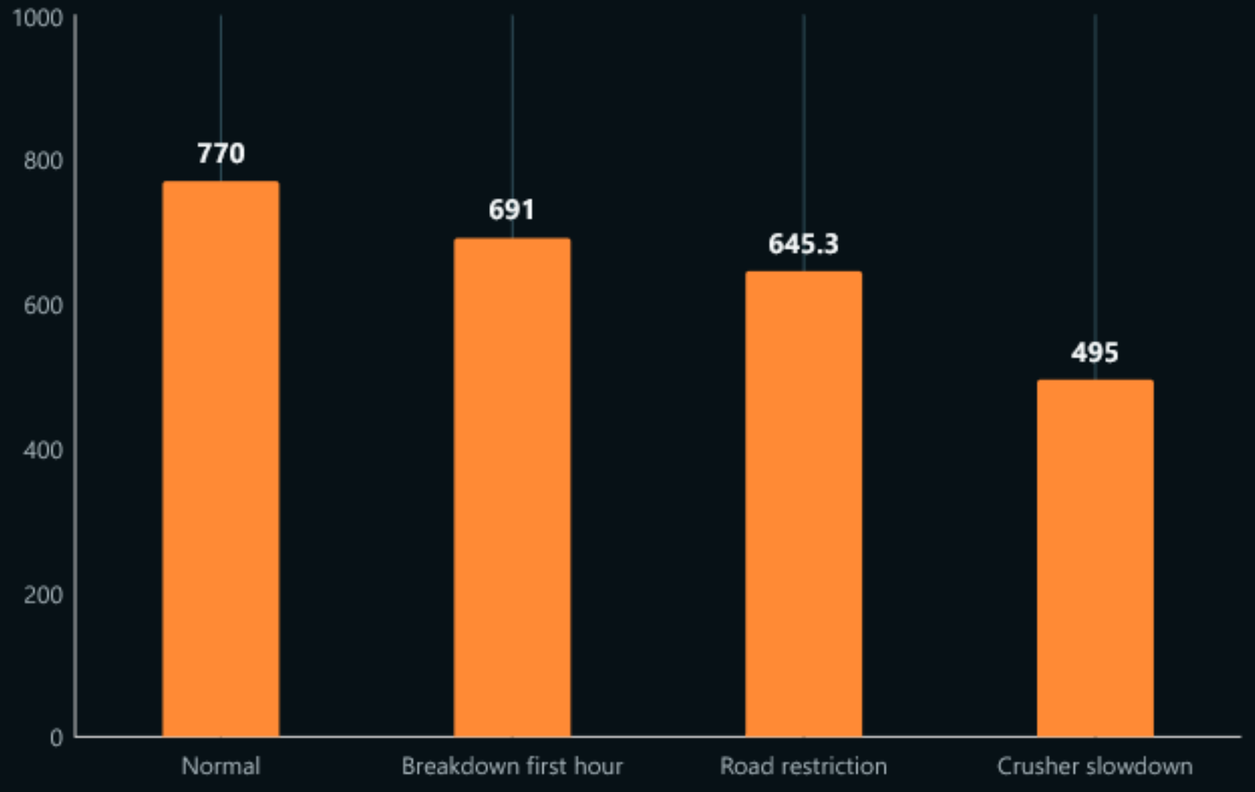
payload in trucks + current ROM + processed ore

The equality is checked throughout every scenario.

Evidence classes

Sourced	OpenMines parameters and licenses
Assumed	Route, fleet and crusher planning values
Synthetic	Telemetry and modeled outcomes

Each disruption changes a visible constraint





Breakdown is the matched returning-lane first-hour case. Other values use the documented steady comparison window.

Smart Dispatch removes avoidable truck activity



4 + 2

cycling + ready reserve

-33%

truck-hours per 1,000 t

-26%

estimated fuel in matched model

Baseline throughput remains 770 t/h. Results are synthetic model evidence, not measured mine savings.

A judge can follow the evidence in three minutes



Judge route

1 Baseline mine

2 HT-01 breakdown

3 Show effect

4 **Decision evidence**

5 Why trust this?

6 Overview and reset

http://localhost:4173

Scope: decision demonstration, not an engineering design, safety system or site forecast.